

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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| [Website](#)

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Education

Embry-Riddle Aeronautical University

M.S. in Aviation

Specialization in Aviation Safety Systems

Daytona Beach, FL

Aug 2025–Present

GPA 4.0/4.0

Embry-Riddle Aeronautical University

B.S. in Aeronautical Science

Minor in Air Transportation

Daytona Beach, FL

May 2023

GPA 3.85/4.0

Research Interests

Aviation Safety Analytics; Air Transportation Systems; Trajectory Data Analytics; Terminal and Low-Altitude Airspace Operations

Research Experience

Graduate Researcher

Advisor: Dr. João S. D. Garcia

School of Graduate Studies, College of Aviation

Sep 2025–Present

- ADS-B-based decomposition of approach path stability into sustained and oscillatory deviations
- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data

Graduate Researcher

Advisor: Dr. Ryan Wallace

Center for Advanced Air Mobility

Feb 2026–Present

- LAANC/UASFM-based assessment of shared-airspace exposure between conventional aircraft and UAS operating altitudes
- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[2] Yoo, T., & Garcia, J. S. D. (Under review). Trajectory Analysis of Wind Exposure and Approach Stability in Flight Training Operations.

[1] Yoo, T., & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. *AIAA AVIATION 2026 Forum*. American Institute of Aeronautics and Astronautics.

Technical Reports

Rodrigues, E., Yoo, T., Pleffken, D. R., Cerqueira, C. S., Racharla, C. V., Crouse, S., Mendonca, F. A. C., Wadler, J., Adkins, K., Ehrenkauf, T., Keebler, J., & Garcia, J. S. D. (2026). *Synthesis framework: A practical tool linking use case features to trust and acceptance outcomes, to inform future pilot projects and public engagement strategies*.

Research Presentations

INFORMS Annual Meeting (Oral) – Accepted	November 2026
Institute for Operations Research and the Management Sciences	San Francisco, CA
<i>From Wind Exposure to Go-Arounds: The Mediating Role of Approach Stability in Commercial Aviation</i>	
UAA 78th Collegiate Aviation Education Conference (Oral)	September 2026
University Aviation Association	Denton, TX
<i>The Effect of Wind Exposure on Go-Around Initiation and the Mediating Role of Approach Stability in Flight Operations</i>	
AIAA AVIATION 2026 Forum (Oral)	June 2026
American Institute of Aeronautics and Astronautics	San Diego, CA
<i>Data-Driven Framework for Assessing Wind Impacts on Approach Stability</i>	
Discovery Day (Poster)	April 2026
Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Characterizing Air Traffic Density Using Nationwide ADS-B Data</i>	
COA Fall Research Colloquium (Oral)	November 2025
College of Aviation, Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Approach Stability Assessment Using ADS-B Data in Flight Training</i>	

Funded Research Projects

2026 Safety Research Seed Grant Program	Boeing Center for Aviation and Aerospace Safety
Graduate Research Lead PI: João S. D. Garcia	Jul 2026–Jun 2027
<i>An Empirical Analysis of Go-Around and Collision Risk in Support of In-Time Aviation Safety Management Systems</i>	

Research Activities Supporting AAM National Strategies	Florida Department of Transportation
Graduate Research Assistant PI: João S. D. Garcia	Feb–Jun 2026
<i>Advanced Air Mobility in Disaster Response and Other Functions of Government as a Driver for Public Acceptance</i>	

Teaching Experience

Graduate Teaching Assistant	Embry-Riddle Aeronautical University
Supervisor: Dr. Dahai Liu	Dec 2025–Present

Teaching support for:

- Statistical Analysis for Aviation/Aerospace
- Research Methods in Aviation/Aerospace
- Graduate Capstone Research Project
- Thesis

Flight Instructor

Instructional areas include:

Aviators Academy
Nov 2023–Jul 2025

- Private Pilot Certificate
- Instrument Rating
- Commercial Pilot Certificate
- Certified Flight Instructor
- Certified Flight Instructor-Instrument
- Flight Review and Instrument Proficiency Check
- Complex Endorsement

Honors & Awards

3 rd Place, Student Research Poster Contest, University Aviation Association	September 2026
Best Poster Award, Embry-Riddle Aeronautical University	April 2026
1 st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University	November 2025
Magna Cum Laude, Embry-Riddle Aeronautical University	May 2023
Dean's List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University	

Certifications & Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane
Flight Instructor – ASEL, AMEL, Instrument Airplane
Ground Instructor – Advanced, Instrument
Remote Pilot – sUAS

Technical Skills

Python, SPSS, LaTeX

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS)
American Institute of Aeronautics and Astronautics (AIAA)
University Aviation Association (UAA)
Royal Aeronautical Society (RAeS)

Leadership & Service

Mentor	American Institute of Aeronautics and Astronautics
AIAA Cape Canaveral Section Mentorship Program	2026–Present
Secretary	Embry-Riddle Aeronautical University
Korean Student Association	2019–2020
Administrative Specialist; Squad Commander	Korea Defense Communication Command
Republic of Korea Military Service	2017–2019
Vice Captain	University of North Dakota
Table Tennis Club	2016–2017